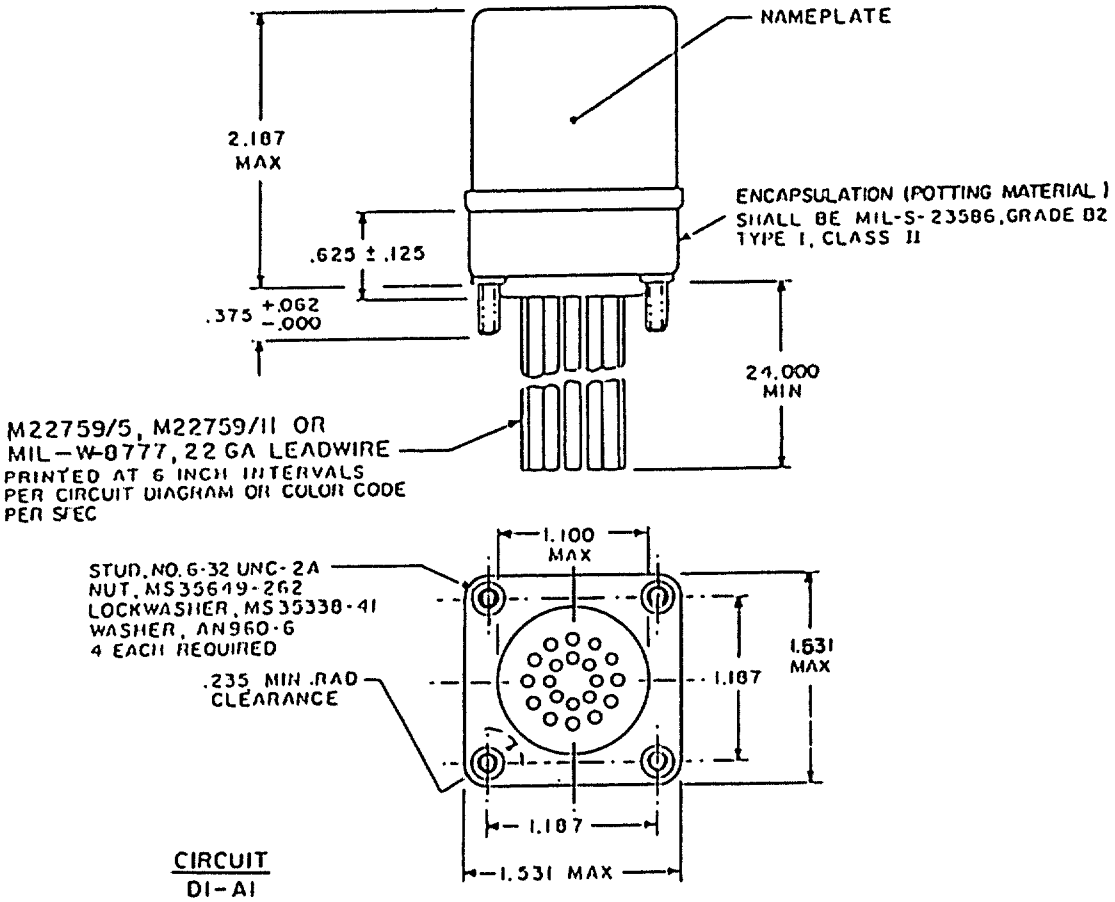
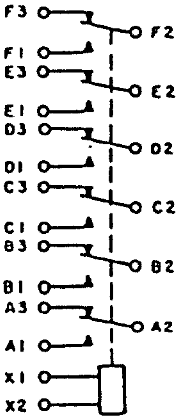
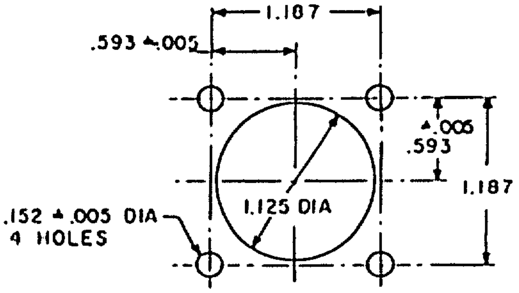
FED. SUP CLASS
5945

~~INACTIVE FOR NEW DESIGN AFTER 5 JUN 87~~
NO SUPERSEDING STANDARD



CIRCUIT
D1-A1
(NOTE 5)





MOUNTING HOLE LAYOUT

(J) denotes changes

APPROVED 17 March 1958 REVISED (J) 20 Jan 1989

User activities: Army - Navy - Air Force -

Review activities: Army - EC Navy - Air Force - 11, 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A
USAF - 85
Other Cust
Navy - AS

International interest

TITLE
RELAYS, ELECTROMAGNETIC, 5 AMPERES,
6 PDT, TYPE I, POTTED LEAD,
HERMETICALLY SEALED

MILITARY STANDARD

MS25270

Procurement Specification
MIL-R-6106

SUPERSEDES:

PAGE 1 OF 5

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

5945-0792-02

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FED. SUP CLASS
5945

Inches	mm
.005	0.13
.062	1.57
.125	3.18
.152	3.86
.235	5.97
.375	9.53
.593	15.06
.025	15.88
1.100	27.94
1.187	30.15
1.531	38.89
2.187	55.55
24.00	609.60

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified hereon.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS25270-	Type	Coil	Terminal type	Mounting	Max weight in pounds
D1	I	dc	Lead	Stud	.67
A1	I	ac	Lead	Stud	.67

P.A USAF - 85 Other Cust Navy - AS	International interest	TITLE RELAYS, ELECTROMAGNETIC, 5 AMPERES, 6 PDT, TYPE I, POTTED LEAD, HERMETICALLY SEALED	MILITARY STANDARD MS25270
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 2 OF 5

DD FORM 1 MAY 73 672 *(Coordinated)* PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

APPROVED 17 March 1958 REVISED (J) For changes see pages 3 and 5

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Review activities: Army - EC
Navy -
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

FED. SUP CLASS
5945

Ⓙ TABLE II. Operating characteristics.

MS part no. MS25270-	Coil data											Time (milliseconds max)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce			
														Main		Aux	
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
01	X1, X2	28	dc	N/A	29	0.13	18	19.8	22.5	7.0	1.5	20	20	2	2		
A1	X1, X2	115	400 5/	N/A	122	0.04	90	95	103	30	5.0	25	50	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25270-A1 may be used on 60 Hz if maximum ambient temperature is +35°C (maximum coil current shall be 0.044 ampere).

APPROVED 17 March 1958 REVISED Ⓙ For changes see pages 3 and 5

P.A USAF - 85
Other Cust
Navy - AS

International interest

TITLE
RELAYS, ELECTROMAGNETIC, 5 AMPERES,
6 PDT, TYPE I, POTTED LEAD,
HERMETICALLY SEALED

MILITARY STANDARD

MS25270

Procurement Specification
MIL-R-6106

SUPERSEDES:

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Review activities:
Army -
Navy - EC
Air Force - 11, 99

User activities:
Army -
Navy -
Air Force -

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	5	5			5	4							
Inductive	100													
Inductive	20	3	3			3	2							
Motor	100	1.5	1.5			1.5	1							
Lamp	100	0.8	0.8			0.8	0.6							
Transfer load														2/
Mechanical life reduced current	400	1.25	1.25			1.25	1							
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

FED. SUP CLASS
5945

APPROVED 17 March 1958 **REVISED** Ⓙ For changes see pages 3 and 5

P.A USAF - 85 Other Cust Navy - AS	International interest	TITLE RELAYS, ELECTROMAGNETIC, 5 AMPERES, 6 PDT, TYPE I, POTTED LEAD, HERMETICALLY SEALED	**MILITARY STANDARD** MS 25270
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 4 OF 5

DD FORM 1 MAY 73 672 *(Coordinated)* PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Review activities: Army -
Navy -
Air Force - EC 11, 99

User activities: Army -
Navy -
Air Force -

FED. SUP CLASS
5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	25 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1,500 Hz
Vibration - random	
Applicable specification	N/A
Power spectral density	N/A
RMS G min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,050 V rms	1,000 V rms
Aux contacts		
All other points	1,050 V rms	1,000 V rms

Dielectric strength (altitude)

		80,000 ft
Coil to case		1,000 V rms
Aux contacts		
All other points		1,000 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	20 amperes
Rupture current	25 amperes
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests are not applicable.

Ⓙ Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

APPROVED 17 March 1958 REVISED Ⓙ For changes see pages 3 and 5

P.A
USAF - 85
Other Cust
Navy - AS

International Interest

TITLE

RELAYS, ELECTROMAGNETIC, 5 AMPERES,
6 PDT, TYPE I, POTTED LEAD,
HERMETICALLY SEALED

MILITARY STANDARD

MS25270

Procurement Specification
MIL-R-6106

SUPERSEDES:

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

U.S. Government Printing Office 1989-604-033/1224M